

1. กำหนดตารางความสัมพันธ์ระหว่างค่า x และ y ได้ดังตาราง

จุดที่	x	y
1	2	9.5
2	4	8.0
3	6	10.5
4	8	39.5
5	10	72.5

จงหาค่า $y = f(x)$ เมื่อ $x = 4.5$ ด้วยวิธี

1.1 LINEAR SPLINE พร้อมเขียน code

1.2 QUADRATIC SPLINE พร้อมเขียน code

1.3 CUBIC SPLINE โดยใช้ ขั้นตอนตามนี้หรือเปิดหนังสือ textbook อ่านได้เลยครับ พร้อมเขียน code

18.6.3 Cubic Splines

The objective in cubic **splines** is to derive a third-order polynomial for each interval between knots, as in

$$f_i(x) = a_i x^3 + b_i x^2 + c_i x + d_i \quad (18.35)$$

Thus, for $n + 1$ data points ($i = 0, 1, 2, \dots, n$), there are n intervals and, consequently, $4n$ unknown constants to evaluate. Just as for quadratic **splines**, $4n$ conditions are required to evaluate the unknowns. These are:

1. The function values must be equal at the interior knots ($2n - 2$ conditions).
2. The first and last functions must pass through the end points (2 conditions).

INTERPOLATION

3. The first derivatives at the interior knots must be equal ($n - 1$ conditions).
4. The second derivatives at the interior knots must be equal ($n - 1$ conditions).
5. The second derivatives at the end knots are zero (2 conditions).

The visual interpretation of condition 5 is that the function becomes a straight line at the end knots. Specification of such an end condition leads to what is termed a "natural" **spline**. It is given this name because the drafting **spline** naturally behaves in this fashion (Fig. 18.15). If the value of the second derivative at the end knots is nonzero (that is, there is some curvature), this information can be used alternatively to supply the two final conditions.

The above five types of conditions provide the total of $4n$ equations required to solve for the $4n$ coefficients. Whereas it is certainly possible to develop cubic **splines** in this fashion, we will present an alternative technique that requires the solution of only $n - 1$ equations. Although the derivation of this method (Box 18.3) is somewhat less straightforward than that for quadratic **splines**, the gain in efficiency is well worth the effort.

แบบฝึกหัด LEAST-SQUARES REGRESSION

1. LINEAR REGRESSION

x	10	15	20	30	40	50	60	70	80
f(x)	5	9	15	18	22	30	35	38	43

จงใช้ LINEAR REGRESSION ในการคำนวณหา $f(x) = a_0 + a_1x_1$ และค่า $f(65)$

- แสดงวิธีทำพร้อมเขียน code

2. POLYNOMIAL REGRESSION

x	10	15	20	30	40	50	60	70	80
f(x)	5	9	15	18	22	30	35	38	43

จงใช้ POLYNOMIAL REGRESSION order $m = 2$ ในการสร้างสมการ $f(x) = a_0 + a_1x + a_2x^2$

- แสดงวิธีทำพร้อมเขียน code